

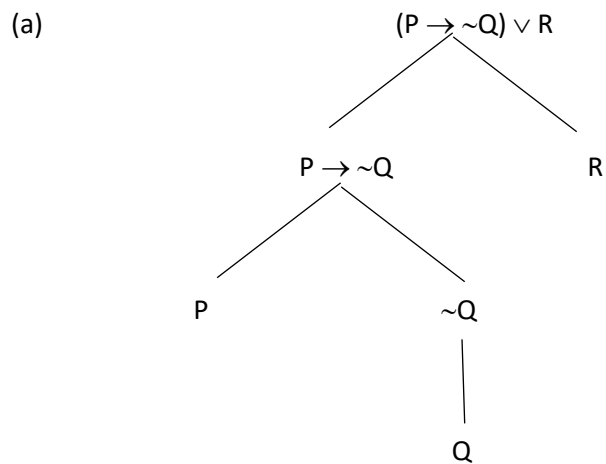
Logic 1

Midterm Answers

Questions

1. Consider the sentence ' $(P \rightarrow \sim Q) \vee R$ '.
 - (a) Analyse this sentence into its components using a tree diagram. (6 marks)
 - (b) What is the main operator of the sentence? (3 marks)

Answer



(b) $\vee$

2. Write down a truth table for the sentence ' $P \rightarrow (P \wedge \sim Q)$ ' (6 marks)

Answer

$P \rightarrow (P \wedge \sim Q)$					
T	F	T	F	F	T
F	T	F	F	F	T
T	T	T	T	T	F
F	T	F	F	T	F

3. Add brackets to make the following truth tables correct, and indicate the main operator:

P	$\wedge$	Q	$\rightarrow$	P
T	T	T	T	T
F	F	T	F	F
T	T	F	T	T
F	F	F	T	F

(5 marks)

$\sim$	P	$\wedge$	Q	$\vee$	R
F	T	T	T	T	T
T	F	F	T	T	T
T	T	F	F	T	T
T	F	F	F	T	T
F	T	T	T	F	F
T	F	F	T	T	F
T	T	F	F	T	F
T	F	F	F	T	F

(5 marks)

Answer

$P \wedge (Q \rightarrow P)$ main operator: $\wedge$

$\sim(P \wedge Q) \vee R$ main operator: $\vee$

4. The more distinct letters there are in a molecular sentence, the *longer* the truth table for that sentence would need to be. So the truth table for ' $P \rightarrow (P \wedge Q)$ ' would need to have four rows, while the truth table for ' $(P \wedge Q) \rightarrow \sim R$ ' would need to have eight rows etc.

(a) Give a general formula for the number of rows in a truth table for a sentence with n distinct letters. (5 marks)

In the lectures we discussed how some philosophers and logicians have posited *additional* truth values, beyond just the values *true* and *false*. Some have suggested, for instance, that a sentence can be *neither* true nor false, which would give us a third truth value. And we could, in principle, posit even more truth values than this.

(b) Give a general formula for the number of rows in a truth table with m distinct truth values for a sentence with n distinct sentence letters. (10 marks)

Answer

(a) 2^n

(b) m^n

5. Translate the following argument into our formal language and construct a derivation to show that it's valid (10 marks):

The kettle will boil only if it's plugged in and switched on.

The kettle is not plugged in.

Therefore, the kettle won't boil.

Answer

P: The kettle will boil.

Q: The kettle is plugged in.

R: The kettle is switched on.

$$P \rightarrow (Q \wedge R)$$

$$\sim Q$$

$$\therefore \sim P$$

1. ~~Show~~ $\sim P$

2.	P	ass id
3.	$P \rightarrow (Q \wedge R)$	pr
4.	$\sim Q$	pr
5.	$Q \wedge R$	2 3 mp
6.	Q	5 s
7.		4 6 id

6. Construct a derivation to show that the following argument is valid (15 marks):

$$(P \rightarrow Q) \rightarrow (Q \rightarrow P)$$

$$\sim(Q \rightarrow P) \vee \sim R$$

$$\sim S \vee R$$

$$\sim S \rightarrow S$$

$$\therefore \sim(P \rightarrow Q)$$

Answer

Here's one possible derivation.

1. ~~Show~~ $\sim(P \rightarrow Q)$

2.	$(P \rightarrow Q) \rightarrow (Q \rightarrow P)$	pr
3.	$\sim(Q \rightarrow P) \vee \sim R$	pr
4.	$\sim S \vee R$	pr
5.	$\sim S \rightarrow S$	pr
6.	Show S	
7.	$\sim S$	ass id
8.	S	5 7 mp
9.		7 8 id
10.	$\sim\sim S$	6 dn
11.	R	4 10 mtp
12.	$\sim\sim R$	11 dn
13.	$\sim(Q \rightarrow P)$	3 12 mtp
14.	$\sim(P \rightarrow Q)$	2 13 mt dd

Or... (this alternate strategy was brought up in the lecture)

1.	Show $\sim(P \rightarrow Q)$	
2.	$P \rightarrow Q$	ass id
3.	$(P \rightarrow Q) \rightarrow (Q \rightarrow P)$	pr
4.	$\sim(Q \rightarrow P) \vee \sim R$	pr
5.	$\sim S \vee R$	pr
6.	$\sim S \rightarrow S$	pr
7.	$Q \rightarrow P$	2 3 mp
8.	$\sim\sim(Q \rightarrow P)$	7 dn
9.	$\sim R$	4 8 mtp
10.	$\sim S$	5 9 mtp
11.	S	6 10 mp
12.		10 11 id

7. A set of sentences is *inconsistent* if and only if it is impossible for them to all be true. The following is an example of an inconsistent set of sentences:

$$\begin{aligned}
 &P \wedge Q \\
 &\sim Q \vee R \\
 &R \rightarrow \sim P
 \end{aligned}$$

- (a) Demonstrate that these sentences are inconsistent using a truth table. (10 marks)
 (b) Demonstrate that these sentences are inconsistent using a derivation. (15 marks)

Answer

			
(a)	$P \wedge Q$	$\sim Q \vee R$	$R \rightarrow \sim P$
	T T T	F T T T	T F FT
	F F T	F T T T	T T TF
	T F F	T F T T	T F FT
	F F F	T F T T	T T TF
	T T T	F T F F	F T FT
	F F T	F T F F	F T TF
	T F F	T F T F	F T FT
	F F F	T F T F	F T TF

There are no rows in which all of the sentences are true.

(b)

There are a few ways to use a derivation to demonstrate that a set of sentences is inconsistent. Remember what it means for a set of sentences to be inconsistent – it is impossible for them all to be true. But then the negation of the conjunction of these sentences would have to be a logical truth. In this case, that would be:

$$\sim((P \wedge Q) \wedge (\sim Q \vee R) \wedge (R \rightarrow \sim P))$$

1.	Show $\sim((P \wedge Q) \wedge (\sim Q \vee R) \wedge (R \rightarrow \sim P))$	
2.	$(P \wedge Q) \wedge (\sim Q \vee R) \wedge (R \rightarrow \sim P)$	ass id
3.	$(P \wedge Q) \wedge (\sim Q \vee R)$	2 s
4.	$P \wedge Q$	3 s
5.	$\sim Q \vee R$	3 s
6.	$R \rightarrow \sim P$	2 s
7.	P	4 s
8.	Q	4 s
9.	$\sim\sim Q$	7 dn
10.	R	4 8 mtp
11.	$\sim P$	5 9 mp
12.		6 10 id

You can also show that a set of sentences is inconsistent by deriving the negation of one sentence from the remaining sentences. If two of the sentences entail the negation of the third then they cannot all be true together. Here's one example of that in the present case:

1.	Show $\sim(P \wedge Q)$	
2.	$P \wedge Q$	ass id
3.	$\sim Q \vee R$	pr
4.	$R \rightarrow \sim P$	pr
5.	P	2 s
6.	Q	2 s
7.	$\sim\sim Q$	6 dn
8.	R	3 7 mtp
9.	$\sim P$	4 8 mp
10.		5 9 id

Another way to show that a set of sentences is inconsistent is to use them to derive a contradiction $P \wedge \sim P$. The above derivations could easily be adapted to a derivation of this kind.

8. Suppose we introduce a new truth value 'N' meaning *neither* true nor false and suppose we give new truth tables for $\sim$ and $\vee$ as follows:

$\square$	$\sim\square$
T	F
N	N
F	T

$\square \vee \bigcirc$		$\square$		
		T	N	F
$\bigcirc$	T	T	T	T
	N	T	N	N
	F	T	N	F

Would the law of excluded middle $P \vee \sim P$ still be a tautology? Explain your answer. (10 marks)

Answer

Here's the basic answer. If P gets the value N then, by the above truth tables, $\sim P$ also gets the value N and so does $P \vee \sim P$. If a tautology is defined as a sentence that only has Ts in its truth table then $P \vee \sim P$ no longer qualifies as a tautology. That's enough to get the marks. But you can go a bit deeper on this question. When we only have two truth values, a tautology can be defined as a sentence that only has Ts in its truth table but could also be defined as a sentence that has no Fs in its truth table. When we only have two truth values, these definitions are equivalent. But once we introduce a third truth value these are no longer equivalent – and there's a substantial question as to which definition we want to go with. If we go with the second definition, then $P \vee \sim P$ would still be a tautology.